Network Address Translation (NAT)

1 What is NAT?

Network Address Translation (NAT) is a networking technique that allows multiple devices inside a private network to share a single public IP address when communicating with the Internet.

NAT is typically implemented on a **router or firewall** that connects:

- a **private network** (LAN)
- to a **public network** (Internet)

It modifies IP address information in packet headers while traffic is in transit.

2 Why NAT is Needed

2.1 IPv4 Address Exhaustion

IPv4 provides approximately:

$$2^{32} \approx 4.3 \text{ billion IP addresses}$$

Due to the explosive growth of the Internet, this number is insufficient.

2.2 Problems Without NAT

Without NAT:

- Every device would require a **public IP**
- ISPs would quickly run out of addresses
- Network security would be weaker

NAT solves this problem by allowing thousands of devices to use **private IP addresses** internally while sharing a small number of public IPs.

3 Private vs Public IP Addresses

3.1 Private IP Ranges

Private IP addresses are **not routable on the Internet**.

- **Class A:** 10.0.0.0 – 10.255.255.255
- **Class B:** 172.16.0.0 – 172.31.255.255

- **Class C:** 192.168.0.0 – 192.168.255.255

3.2 Public IP Addresses

Public IPs are:

- Globally unique
- Assigned by ISPs
- Routable on the Internet

4 How NAT Works (Step-by-Step)

Scenario:
- PC inside LAN: `192.168.1.10`
- Router public IP: `103.45.12.8`
- Destination server: `8.8.8.8`

4.1 Outgoing Packet

1. PC sends packet from `192.168.1.10` to `8.8.8.8`
2. Router replaces source IP:

$$192.168.1.10 \;\; ! \;\; 103.45.12.8$$

3. Router stores mapping in NAT table

4.2 Incoming Packet

1. Reply arrives at router for `103.45.12.8`
2. Router checks NAT table
3. Packet forwarded to `192.168.1.10`

NAT is **transparent** — internal devices are unaware that translation occurs.

5 Types of NAT

5.1 Static NAT

- One-to-one mapping
- Private IP $ Public IP
- Used for servers